

Group is putting up an end-to-end project to produce GH in Gujarat with a capacity of 1 million metric tons per annum (MTPA). A joint venture has been formed with Kowa, Singapore for sales and marketing of GH and its derivatives. Proximity of Adani's Mundra Port to the global supply chain offers a great opportunity to export GH.

Larsen and Toubro (L&T). The engineering major L&T, in collaboration with the Indian Oil corporation (IOC) and renewable power company ReNew, is investing ₹320 billion, spread over 5 years to produce 45 kg GH/day. It has entered into an agreement with McPhy, the leading electrolyser technology company to manufacture pressurised alkaline electrolyzers in Gujarat.

Other industrial organisations. State-owned NTPC is developing India's first green hydrogen-to-electricity project using solid-oxide electrolyzers. ACME Solar has commissioned the world's first commercial pilot to produce integrated GH and green ammonia in Rajasthan and is installing a solar-to-GH-to-green ammonia unit in Karnataka to produce 1.2MTPA GH. The state-owned refineries Indian Oil Corporation (IOC), Oil India Limited (OIL), Hindustan Petroleum Corporation (HPCL), and Bharat Petroleum



Fig. 9: Dhirubhai Ambani Green Energy Integrated Giga Complex (Source: [auto.economictimes.indiatimes.com](https://economictimes.indiatimes.com))



Fig. 10: Adani New Industries project site (Source: <https://chemindigest.com>)

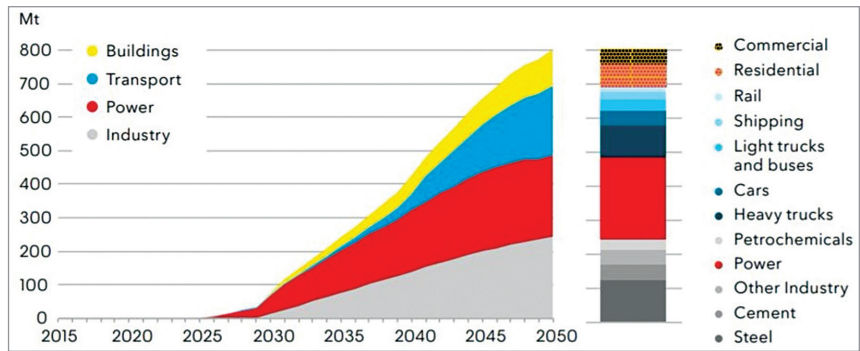


Fig. 11: Global hydrogen demand in the NCS-CEHP (Source: BloombergNEF)

Box 3: Hydrogen internal combustion engines (H2-ICE)

Hydrogen internal combustion engines use hydrogen or a mixture of hydrogen and diesel as the fuel and can run approximately 400km on a single hydrogen fill; the filling taking just about 15 minutes. The conventional internal combustion engines can be converted into H2-ICE engines just by making a few minor modifications.

Corporation (BPCL) are progressively replacing hydrogen with GH in their refining processes. Gujarat Industries Power Company Limited (GIPCL) is setting up a 10MW plant in Gujarat. The Indian Railways is planning to run trains powered by GH and is setting up a plant in Egypt to produce 20,000 tonnes GH annually. The India Hydrogen Alliance (IH2A), an industry-led coalition of global and Indian companies, is planning to put up 25 National Green Hydrogen Projects and 5 National H2 Hubs in the country by 2025 at an investment of US\$468 million. The hubs are to create renewable energy and green ammonia production facilities and provide the infrastructure for storage and evacuation of GH.

Endpoint

Green hydrogen is getting unprecedented momentum and global attention as the green fuel of the future to combat the climate change and global warming issues. It is feasible to use GH in almost every sector which is dependent on fossil fuels and is difficult to decarbonise.

According to a report by BloombergNEF (BNEF), the global hydrogen demand in the NEO Climate Scenario (NCS) - Clean Electricity and Hydrogen Pathway (CEHP) is estimated to

shoot up to about 800 million tonnes by 2050, providing a great investment opportunity. Though GH industry is still in its nascent stage of development, the industry giants are investing heavily to capture the opportunity and accelerate the deployment of GH. They are developing more efficient and cost-effective production technologies, realising the economies of scale, and hastening the development of a robust infrastructure for transportation, distribution, and refuelling.

At present, GH is much more expensive to produce than the grey hydrogen, due to the high initial capital costs of building and operating electrolysis facilities, low efficiency, and high cost of electrolyzers and cost of renewable energy. Reducing these is essential to make GH competitive. Providing financial support, close collaboration amongst countries, and extending the arc of application can help bring down the cost. Sharing the learning, educating the stakeholders, and global cooperation can help increase the user confidence.

India is leading with an ambitious target of producing 5 million metric tonne GH per annum by 2030. Now it is crucial to develop the ecosystem to transition to a green hydrogen economy.

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